Fund. of Data Science Practical Project

Project Description

This project highlights that prediction in data science can be achieved using:

- Statistical methods (Time Series)
- Machine Learning techniques

Both approaches are essential and complementary in real-world decision-making.

Instructions

- Students will work in **groups of 4**.
- Each group is required to complete **BOTH** two projects.
- Each group must select **unique datasets and application domains** (no duplication between groups).
- Due date: week 14

Deliverables

Each group must submit:

- Source code for both projects
- A written report explaining problem, dataset, methodology, results, and conclusion

---

Project 1: Time Series Forecasting Project

Students will use historical time-based data to predict future values using simple forecasting techniques.

📊 Possible Datasets

- Temperature over time
- Stock prices
- Patient vital signs (heart rate / glucose)
- Website traffic

⚙️ Required Tasks

- Load dataset (CSV format)
- Visualize data using line plots
- Apply:
  - Simple Moving Average (SMA)
  - Exponential Moving Average (EMA)
- Forecast future values
- Compare results between methods

📈 Evaluation

- Visual comparison of results (plots)
  Error measurement (optional, e.g., MAE)

Project 2: Building a Predictive Model for Decision Making using ML

Students will build a machine learning model to predict an outcome (classification or regression) based on a selected dataset.

📊 Possible Datasets

- Diabetes dataset
- Heart disease dataset
- Customer churn
- Student performance

⚙️ Required Tasks

- Perform data preprocessing
- Select features and target variable
- Apply one ML model
- Train and test the model
- Evaluate performance

📈 Evaluation

- Accuracy or RMSE
- Confusion matrix (for classification problems)

Project 1: Time Series Sample Datasets
1. Daily Minimum Temperature
https://raw.githubusercontent.com/jbrownlee/Datasets/master/daily-min-temperatures.csv
2. Airline Passengers
https://raw.githubusercontent.com/jbrownlee/Datasets/master/airline-passengers.csv
3. Monthly Sales / Traffic Data
https://raw.githubusercontent.com/selva86/datasets/master/a10.csv

---

Project 2: Machine Learning Sample Datasets

1. Diabetes Dataset

https://raw.githubusercontent.com/plotly/datasets/master/diabetes.csv

2. Heart Disease Dataset

https://raw.githubusercontent.com/dataprofessor/data/master/heart_disease.csv
3. Student Dataset (Tips)
https://raw.githubusercontent.com/mwaskom/seaborn-data/master/tips.csv